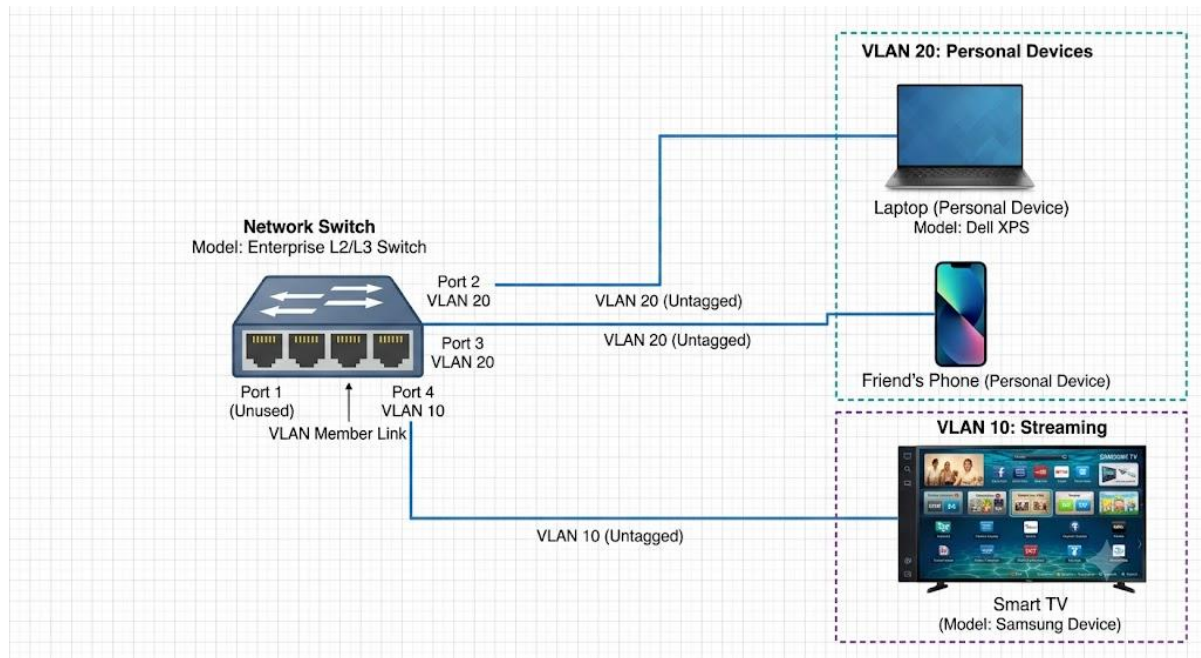


SWITCHING AND VLANS

Task – 1 Labelling in the devices with the VLANS



Task – 2 Simulate with cisco tracker.

- Here is the easiest way to configure and test this using the point-and-click Graphical User Interface (GUI) in Cisco Packet Tracer, bypassing the command line entirely.
-
- 1. Set Up the Devices
- Drag one Switch (e.g., 2960) and five PCs into your workspace.
- Connect the PCs to the switch using copper straight-through cables (the solid black line).
-
- 2. Assign IP Addresses (GUI)
- Click a PC, go to the Desktop tab, and select IP Configuration.

- Give three PCs IPs for the first network: 192.168.10.1, 192.168.10.2, and 192.168.10.3.
- Give the other two PCs IPs for the second network: 192.168.20.1 and 192.168.20.2.
- Set the Subnet Mask to 255.255.255.0 for all PCs.
-
- 3. Create VLANs on the Switch (GUI)
- Click the Switch and go to the Config tab.
- On the left menu, click VLAN Database.
- Type 10 in the VLAN Number box, name it VLAN10, and click Add.
- Type 20 in the VLAN Number box, name it VLAN20, and click Add.
-
- 4. Assign Ports to VLANs (GUI)
- Still in the switch Config tab, look at the left menu under "Interfaces".
- Click FastEthernet0/1 (the port connecting your first PC).
- Find the VLAN dropdown menu and change it from 1 to 10.
- Repeat this for all your connected ports: assign the first three interfaces to VLAN 10, and the other two to VLAN 20.
-
- 5. Test the Connection
- Look at the right-side toolbar and click the Closed Envelope icon (Add Simple PDU).
- Click one PC in VLAN 10, then click another PC in VLAN 10. The status window in the bottom right will say Successful.
- Click a PC in VLAN 10, then click a PC in VLAN 20. The status will say Failed, proving your VLANs are successfully blocking traffic between the two different groups.

Task – 3 Switching the VLAN 10 to VLAN 20

- Moving a PC from one VLAN to another means changing which group of devices it is allowed to talk to on the switch.
-
- Step 1: The "Before" Setup (PC is in VLAN 10)
- Switch Command: You put port FastEthernet 0/1 into VLAN 10.
- Plaintext
- Switch(config)# interface FastEthernet 0/1
- Switch(config-if)# switchport access vlan 10

- Set the PC's IP to match VLAN 10 (e.g., 192.168.10.50).
 - Test:
 - Ping a device in VLAN 10 (192.168.10.10): Works!
 - Ping a device in VLAN 20 (192.168.20.10): Fails.
 -
 - Step 2: Move the PC (Changing to VLAN 20)
 - Switch Command: Type the exact same command, but change 10 to 20. The switch automatically replaces the old setting.
- ```

- Plaintext
- Switch(config)# interface FastEthernet 0/1
 Switch(config-if)# switchport access vlan 20

```
- Change the PC's IP to match the new network in VLAN 20 (e.g., 192.168.20.50).
  - 
  - Step 3: The "After" Test
    - Open the PC's Command Prompt and try pinging again:
      - Ping a device in VLAN 20 (192.168.20.10): Works!
      - Ping a device in VLAN 10 (192.168.10.10): Fails.
    - The switch now tags all traffic coming from that PC with VLAN 20, so it can only talk to other devices inside VLAN 20 and is completely cut off from VLAN 10.

## Task – 4 Assigning the 10 ports on or with VLANS 30.

- Here is the simplest Cisco IOS script using the interface range command (supported in Packet Tracer, NetSim, and GNS3) to configure ports 1 through 10 in one single step.

```

- Plaintext
- Switch# configure terminal
 Switch(config)# vlan 30
 Switch(config-vlan)# name Automated_VLAN
 Switch(config-vlan)# exit
 Switch(config)# interface range FastEthernet 0/1 - 10
 Switch(config-if-range)# switchport mode access
 Switch(config-if-range)# switchport access vlan 30
 Switch(config-if-range)# no shutdown
 Switch(config-if-range)# end

```

## Switch# write memory

- vlan 30 & name Automated\_VLAN: Creates VLAN 30 on the switch and gives it a label.
- interface range FastEthernet 0/1 - 10: **The shortcut.** Applies all following commands simultaneously to ports Fa0/1 through Fa0/10 instead of doing them one by one.
- switchport mode access: Sets all 10 ports to access mode (for end-user devices like PCs or printers).
- switchport access vlan 30: Assigns all 10 ports to VLAN 30 at once.
- no shutdown: Ensures all 10 interfaces are turned on and active.
- write memory: Saves the configuration so it isn't lost if the switch reboots.
- Type show vlan brief on the switch CLI to confirm the change. Ports Fa0/1 through Fa0/10 will all appear under VLAN 30.
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